and similarly for the conditional PDF $p_{Y|X}$. Bayes' rule expresses the conditional PDF $p_{Y|X}$ with $p_{X|Y}$ by

$$p_{Y|X}(y|x) = \frac{p_{X|Y}(x|y)p_Y(y)}{p_X(x)}, \tag{102}$$

for $p_X(x) > 0$.

The *conditional expectation* $\mathbb{E}[X|Y]$ is the best approximating *function* $g_\star(Y)$ to X in the least-squares sense:

$$\begin{aligned} g_\star &\coloneqq \operatorname*{arg\,min}_{g:\mathbb{R}^d\to\mathbb{R}^d} \mathbb{E}\left[\|X - g(Y)\|^2\right] = \operatorname*{arg\,min}_{g:\mathbb{R}^d\to\mathbb{R}^d} \int \|x - g(y)\|^2 p_{X,Y}(x,y)\mathrm{d}x\mathrm{d}y \\ &= \operatorname*{arg\,min}_{g:\mathbb{R}^d\to\mathbb{R}^d} \int \left[\int \|x - g(y)\|^2 p_{X|Y}(x|y)\mathrm{d}x\right] p_Y(y)\mathrm{d}y. \end{aligned} \tag{103}$$

For $y \in \mathbb{R}^d$ such that $p_Y(y) > 0$ the conditional expectation function is therefore

$$\mathbb{E}[X|Y = y] \coloneqq g_\star(y) = \int x p_{X|Y}(x|y)\mathrm{d}x, \tag{104}$$

where the second equality follows from taking the minimizer of the inner brackets in Equation (103) for $Y = y$, similarly to Equation (98). Composing $g_\star$ with the random variable Y, we get

$$\mathbb{E}[X|Y] \coloneqq g_\star(Y), \tag{105}$$

which is a random variable in $\mathbb{R}^d$. Rather confusingly, both $\mathbb{E}[X|Y = y]$ and $\mathbb{E}[X|Y]$ are often called *conditional expectation*, but these are different objects. In particular, $\mathbb{E}[X|Y = y]$ is a function $\mathbb{R}^d \to \mathbb{R}^d$, while $\mathbb{E}[X|Y]$ is a random variable assuming values in $\mathbb{R}^d$. To disambiguate these two terms, our discussions will employ the notations introduced here.

The *tower property* is an useful property that helps simplify derivations involving conditional expectations of two RVs X and Y:

$$\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X] \tag{106}$$

Because $\mathbb{E}[X|Y]$ is a RV, itself a function of the RV Y, the outer expectation computes the expectation of $\mathbb{E}[X|Y]$. The tower property can be verified by using some of the definitions above:

$$\begin{aligned} \mathbb{E}[\mathbb{E}[X|Y]] &= \int \left(\int x p_{X|Y}(x|y)\mathrm{d}x\right) p_Y(y)\mathrm{d}y \\ &\overset{(101)}{=} \int\int x p_{X,Y}(x,y)\mathrm{d}x\mathrm{d}y \\ &\overset{(100)}{=} \int x p_X(x)\mathrm{d}x = \mathbb{E}[X]. \end{aligned}$$

Finally, consider a helpful property involving two RVs $f(X,Y)$ and Y, where X and Y are two arbitrary RVs. Then, by using the Law of the Unconscious Statistician with (104), we obtain the identity

$$\mathbb{E}[f(X,Y)|Y = y] = \int f(x,y) p_{X|Y}(x|y)\mathrm{d}x. \tag{107}$$

B A Proof of the Fokker-Planck equation

In this section, we give here a self-contained proof of the Fokker-Planck equation which includes the continuity equation as a special case (Theorem 11). We stress that **this section is not necessary to understand the remainder of this document** and is mathematically more advanced. If you desire to understand where the Fokker-Planck equation comes from, then this section is for you.

Theorem 41 (Fokker-Planck Equation)
Let p_t be a probability path with $p_0 = p_{\text{init}}$ and let us consider the SDE

$$X_0 \sim p_{\text{init}}, \quad \mathrm{d}X_t = u_t(X_t)\mathrm{d}t + \sigma_t \mathrm{d}W_t.$$

Then X_t has distribution p_t for all $0 \leq t \leq 1$ if and only if the **Fokker-Planck equation** holds:

$$\partial_t p_t(x) = -\mathrm{div}(p_t u_t)(x) + \frac{\sigma_t^2}{2}\Delta p_t(x) \quad \text{for all } x \in \mathbb{R}^d, 0 \leq t \leq 1, \tag{108}$$

We start by showing that the Fokker-Planck is a necessary condition, i.e. if $X_t \sim p_t$, then the Fokker-Planck equation is fulfilled. The trick for the proof is to use **test functions** f, i.e. functions $f : \mathbb{R}^d \to \mathbb{R}$ that are infinitely differentiable ("smooth") and are only non-zero within a bounded domain (compact support). We use the fact that for arbitrary integrable functions $g_1, g_2 : \mathbb{R}^d \to \mathbb{R}$ it holds that

$$g_1(x) = g_2(x) \text{ for all } x \in \mathbb{R}^d \quad \Leftrightarrow \quad \int f(x) g_1(x) \mathrm{d}x = \int f(x) g_2(x) \mathrm{d}x \text{ for all test functions } f \tag{109}$$

In other words, we can express the pointwise equality as equality of taking integrals. The useful thing about test functions is that they are smooth, i.e. we can take gradients and higher-order derivatives. In particular, we can use **integration by parts** for arbitrary test functions f_1, f_2:

$$\int f_1(x) \frac{\partial}{\partial x_i} f_2(x) \mathrm{d}x = -\int f_2(x) \frac{\partial}{\partial x_i} f_1(x) \mathrm{d}x \tag{110}$$

under the condition that f_1, f_2 and their product $f_1 \cdot f_2$ is integrable. By using this together with the definition of the divergence and Laplacian (see Equation (22)), we get the identities:

$$\int \nabla f_1^T(x) f_2(x) \mathrm{d}x = -\int f_1(x) \mathrm{div}(f_2)(x) \mathrm{d}x \quad (f_1 : \mathbb{R}^d \to \mathbb{R}, f_2 : \mathbb{R}^d \to \mathbb{R}^d) \tag{111}$$

$$\int f_1(x) \Delta f_2(x) \mathrm{d}x = \int f_2(x) \Delta f_1(x) \mathrm{d}x \quad (f_1 : \mathbb{R}^d \to \mathbb{R}, f_2 : \mathbb{R}^d \to \mathbb{R}) \tag{112}$$

Now let's proceed to the proof. We use the stochastic update of SDE trajectories as in Equation (6):

$$X_{t+h} = X_t + h u_t(X_t) + \sigma_t(W_{t+h} - W_t) + h R_t(h) \tag{113}$$

$$\approx X_t + h u_t(X_t) + \sigma_t(W_{t+h} - W_t) \tag{114}$$

where for now we simply ignore the error term $R_t(h)$ for readability as we will take $h \to 0$ anyway. We can then

make the following calculation:

$$\begin{aligned}
f(X_{t+h}) - f(X_t) &\overset{(114)}{=} f(X_t + hu_t(X_t) + \sigma_t(W_{t+h} - W_t)) - f(X_t) \\
&\overset{(i)}{=} \nabla f(X_t)^T \left(hu_t(X_t) + \sigma_t(W_{t+h} - W_t)\right) \\
&\quad + \frac{1}{2}\left(hu_t(X_t) + \sigma_t(W_{t+h} - W_t)\right)^T \nabla^2 f(X_t) \left(hu_t(X_t) + \sigma_t(W_{t+h} - W_t)\right) \\
&\overset{(ii)}{=} h\nabla f(X_t)^T u_t(X_t) + \sigma_t \nabla f(X_t)^T (W_{t+h} - W_t) \\
&\quad + \frac{1}{2} h^2 u_t(X_t)^T \nabla^2 f(X_t) u_t(X_t) + h\sigma_t u_t(X_t)^T \nabla^2 f(X_t)(W_{t+h} - W_t) + \\
&\quad + \frac{1}{2}\sigma_t^2 (W_{t+h} - W_t)^T \nabla^2 f(X_t)(W_{t+h} - W_t)
\end{aligned}$$

where in (i) we used a 2nd Taylor approximation of f around X_t and in (ii) we used the fact that the Hessian $\nabla^2 f$ is a symmetric matrix. Note that $\mathbb{E}[W_{t+h} - W_t | X_t] = 0$ and $W_{t+h} - W_t | X_t \sim \mathcal{N}(0, hI_d)$. Therefore

$$\begin{aligned}
&\mathbb{E}[f(X_{t+h}) - f(X_t)|X_t] \\
=&h\nabla f(X_t)^T u_t(X_t) + \frac{1}{2} h^2 u_t(X_t)^T \nabla^2 f(X_t) u_t(X_t) + \frac{h}{2}\sigma_t^2 \mathbb{E}_{\epsilon_t \sim \mathcal{N}(0, I_d)}[\epsilon_t^T \nabla^2 f(X_t) \epsilon_t] \\
\overset{(i)}{=}&h\nabla f(X_t)^T u_t(X_t) + \frac{1}{2} h^2 u_t(X_t)^T \nabla^2 f(X_t) u_t(X_t) + \frac{h}{2}\sigma_t^2 \mathrm{trace}(\nabla^2 f(X_t)) \\
\overset{(ii)}{=}&h\nabla f(X_t)^T u_t(X_t) + \frac{1}{2} h^2 u_t(X_t)^T \nabla^2 f(X_t) u_t(X_t) + \frac{h}{2}\sigma_t^2 \Delta f(X_t)
\end{aligned}$$

where in (i) we used the fact that $\mathbb{E}_{\epsilon_t \sim \mathcal{N}(0, I_d)}[\epsilon_t^T A \epsilon_t] = \mathrm{trace}(A)$ and in (ii) we used the definition of the Laplacian and the Hessian matrix. With this, we get that

$$\begin{aligned}
&\partial_t \mathbb{E}[f(X_t)] \\
=&\lim_{h \to 0} \frac{1}{h} \mathbb{E}[f(X_{t+h}) - f(X_t)] \\
=&\lim_{h \to 0} \frac{1}{h} \mathbb{E}[\mathbb{E}[f(X_{t+h}) - f(X_t)|X_t]] \\
=&\mathbb{E}[\lim_{h \to 0} \frac{1}{h} \left(h\nabla f(X_t)^T u_t(X_t) + \frac{1}{2} h^2 u_t(X_t)^T \nabla^2 f(X_t) u_t(X_t) + \frac{h}{2}\sigma_t^2 \Delta f(X_t)\right)] \\
=&\mathbb{E}[\nabla f(X_t)^T u_t(X_t) + \frac{1}{2}\sigma_t^2 \Delta f(X_t)] \\
\overset{(i)}{=}&\int \nabla f(x)^T u_t(x) p_t(x) \mathrm{d}x + \int \frac{1}{2}\sigma_t^2 \Delta f(x) p_t(x) \mathrm{d}x \\
\overset{(ii)}{=}&-\int f(x) \mathrm{div}(u_t p_t)(x) \mathrm{d}x + \int \frac{1}{2}\sigma_t^2 f(x) \Delta p_t(x) \mathrm{d}x \\
=&\int f(x) \left(-\mathrm{div}(u_t p_t)(x) + \frac{1}{2}\sigma_t^2 \Delta p_t(x)\right) \mathrm{d}x
\end{aligned}$$

where in (i) we used the assumption that p_t as the distribution of X_t and in (ii) we used Equation (111) and Equation (112). Note that to use this, we require integrability of the product $p_t(x)u_t(x)$, i.e. such that

$$\int p_t(x) \|u_t(x)\| \mathrm{d}x < \infty$$

Note that this condition almost always holds in machine learning (bounded data and functions because of numerical precision limits). Therefore, it holds that

$$\partial_t \mathbb{E}[f(X_t)] = \int f(x) \left(-\operatorname{div}(p_t u_t)(x) + \frac{\sigma_t^2}{2} \Delta p_t(x) \right) \mathrm{d}x \quad \text{(for all } f \text{ and } 0 \leq t \leq 1) \tag{115}$$

$$\overset{(i)}{\Leftrightarrow} \quad \partial_t \int f(x) p_t(x) \mathrm{d}x = \int f(x) \left(-\operatorname{div}(p_t u_t)(x) + \frac{\sigma_t^2}{2} \Delta p_t(x) \right) \mathrm{d}x \quad \text{(for all } f \text{ and } 0 \leq t \leq 1) \tag{116}$$

$$\overset{(ii)}{\Leftrightarrow} \quad \int f(x) \partial_t p_t(x) \mathrm{d}x = \int f(x) \left(-\operatorname{div}(p_t u_t)(x) + \frac{\sigma_t^2}{2} \Delta p_t(x) \right) \mathrm{d}x \quad \text{(for all } f \text{ and } 0 \leq t \leq 1) \tag{117}$$

$$\overset{(iii)}{\Leftrightarrow} \quad \partial_t p_t(x) = -\operatorname{div}(p_t u_t)(x) + \frac{\sigma_t^2}{2} \Delta p_t(x) \quad \text{(for all } x \in \mathbb{R}^d, 0 \leq t \leq 1) \tag{118}$$

where in (i) we used the assumption that $X_t \sim p_t$, in (ii) we swapped the derivative with the integral and (iii) we used Equation (109) . This completes the proof that the Fokker-Planck equation is a necessary condition.

Finally, we explain why it is also a sufficient condition. The Fokker-Planck equation is a partial differential equation (PDE). More specifically, it is a so-called *parabolic partial differential equation*. Similar to Theorem 3, such differential equations have a unique solution given fixed initial conditions (see e.g. [15, Chapter 7]). Now, if Equation (108) holds for p_t, we just shown above that it must also hold for true distribution q_t of X_t (i.e. $X_t \sim q_t$) - in other words, both p_t and q_t are solutions to the parabolic PDE. Further, we know that the initial conditions are the same, i.e. $p_0 = q_0 = p_{\text{init}}$ by construction of an interpolating probability path. Hence, by uniqueness of the solution of the differential equation, we know that $p_t = q_t$ for all $0 \leq t \leq 1$ - which means $X_t \sim q_t = p_t$ and which is what we wanted to show.

C Existence and Uniqueness of Continuous-time Markov chains

We prove Theorem 33 in this section.

Proof. **Uniqueness:** We need to show that there can be only one transition kernel $p_{t'|t}(X_{t'} = y | X_t = x)$ that satisfies Equation (87). As a first step, we realize that Equation (87) implies that

$$\frac{\mathrm{d}}{\mathrm{d}t'} p_{t'|t}(X_{t'} = y | X_t = x) \tag{119}$$

$$= \frac{\mathrm{d}}{\mathrm{d}h} p_{t'+h|t}(X_{t'+h} = y | X_t = x)_{|h=0} \tag{120}$$

$$= \frac{\mathrm{d}}{\mathrm{d}h} \left[\sum_{z \in S} p_{t'+h|t'}(X_{t'+h} = y | X_{t'} = z) p_{t'|t}(X_{t'} = z | X_t = x) \right]_{|h=0} \tag{121}$$

$$= \sum_{z \in S} Q_{t'}(y|z) p_{t'|t}(X_{t'} = z | X_t = x) \tag{122}$$

For fixed x, t, one can consider $t' \mapsto p_{t'|t}(X_{t'} = y | X_t = x)$ as a vector-valued function and the above is a linear ODE of that function (the Kolmgorov forward equation, in fact, see Proposition 2) with a known initial condition, i.e. $p_{t|t}(X_t = y | X_t = x) = \delta_y(x)$. As we know, every linear ODE has a unique solution (see Theorem 3), therefore $p_{t'|t}(X_{t'} = y | X_t = x)$ must also be unique.